BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 218 · Dijkstra: shortest paths I

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why does quicksort exclude the pivot from its recursive calls?

Q2. Define a weighted edge.

Q3. What is the first index of an array?

Q4. A–B costs 3 and B–C costs 5. What is cost of A→B→C?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

tentative distance — Best known cost to an unsettled vertex.

settled — Vertex whose shortest distance has been finalised.

relaxation — Improve a neighbour's distance via the current vertex.

predecessor — Previous vertex on the best-known route.

non-negative weight — Edge cost of zero or greater.

New learning

Initialise from the start — Distance(A)=0, all others infinity. Relax A's edges to set B=4 and C=2.

Always choose smallest tentative distance — C=2 settles before B=4. Through C, B improves to 3.

Keep updating neighbours — Settling B changes D from 10 to 8; settling D changes E from 12 to 10.

Reconstruct from predecessors — E←D←B←C←A. Reverse these arrows for A→C→B→D→E, total distance 10.

Dijkstra's requirement — Edge weights must be non-negative. A visited/settled node's shortest distance is then final.

Worked example

Find the shortest P→S route using non-negative weights.

1. Predict the next state — P gives tentative Q=3 and R=7.

2. Reveal and explain — Settle Q=3, improving R to 5 and discovering S=9.

3. Reveal and explain — Settle R=5, improving S to 6.

4. Reveal and explain — Predecessors S←R←Q←P give P→Q→R→S, cost 6.

We Do · Practise with guidance

Find the shortest M→P route using non-negative weights.

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Edges: M-N=5, M-O=2, O-N=1, N-P=3, O-P=8.
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Predict the next state

From M, set N=5 and O=2; O is the next vertex settled.

Reveal and explain

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

From O, N improves from 5 to 3 and P gets a tentative 10.

From N, P improves to 6; P is next. Route M→O→N→P costs 6.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

A–B=4 and A–C=2. Starting Dijkstra at A, state the tentative distances of A, B and C after relaxing A's neighbours.

Question 2

On edges S–A=3, S–B=7, A–B=2, B–T=2 and A–T=8, find the shortest S→T path with full distance reasoning.

Question 3

Explain why Dijkstra must not be relied upon when negative-weight edges are possible, and contrast tentative and settled distances.

Exit Ticket · Answer before checking

Dijkstra always chooses the next vertex by its smallest edge weight.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The pivot is already in its final sorted position.

Q2. A connection between vertices with an associated cost or distance.

Q3. 0.

Q4. 8.

You Do pages · Independent questions

Q1. [3] A=0 [1], B=4 [1], C=2 [1]. C is chosen next.

Q2. [6] After S: A3, B7 [1]; settle A3 [1] improving B to5 and T to11 [2]; settle B5, T improves to7 [1]; route S→A→B→T, cost 7 [1].

Q3. [5] Dijkstra assumes non-negative edges [1], so no later path can improve a settled vertex [1]; a negative edge can invalidate that assumption [1]. Tentative distances can change [1], but settled distances are final under the precondition [1].

Exit Ticket · Model answer

False. It chooses the unsettled vertex with the smallest total tentative distance from the start.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding